

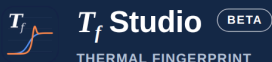


T_f Studio

BETA

THERMAL FINGERPRINT

User Guide



Every glass carries the **thermal fingerprint** of how it was made, written in its fictive temperature. Tf Studio simulates and fits that history, using the heterogeneous **Takeda-Lucas** model alongside **TNM** and **RelaxPy**, for DSC, MDSC, enthalpy, volume and annealing data.

Start learning

Analyze data

Open Fitter

Download for Mac / Windows

TOOLS

Learn EXPERIMENTS & MODELS

How structural relaxation is measured, and what the TNM, MAP and Takeda-Lucas models assume, with live simulations throughout.

- Annealing, Kovacs asymmetry and memory, DSC, enthalpy recovery, MDSC (C_p^* , reversing/non-reversing and their caveats)
- The mathematics of TNM, MAP (RelaxPy) and TL
- Mean field vs Prony vs domain $T_{f,i}$: see the distributions

[Open Learn](#) →

Data Analysis YOUR DATA

Turn raw DSC heat flow and relaxation data into normalized curves, fictive temperatures and enthalpy.

- Glass and liquid baselines → C_p^N , $T_f(T)$, T_f^* for heating and cooling scans
- Heating after annealing: aligned scans, ΔH_F , recovered $\Delta H(t_d)$, ϕ and a KWW fit
- Volume / density / length during annealing: stretched-exponential fit

[Open Data Analysis](#) →

Fitter STEP BY STEP

Program any thermal history, load your measurements, then compute with chosen parameters or fit them.

- Ramps, anneals with log-spaced times, T-jumps, MDSC modulation
- DSC, MDSC C_p/C_p^* , enthalpy, volume, T_f , annealing data
- Bounded local / global fits, errors, correlations, AIC/BIC

[Open Fitter](#) →

Hands-on: the [Lab](#) runs cool-heat scans, anneals and T-jumps with all three models side by side, with relaxation-time distributions and $\beta_{KWW}(T)$.

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2	Getting Tf Studio: browser, Mac and Windows, updates
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5	Lab: quick interactive runs
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In a hurry?

The separate **Quick Start** document takes you from installing to a first result in a few minutes. This guide explains every screen and option in detail.

1 · What Tf Studio does

Every glass carries a thermal fingerprint of how it was made, written in its fictive temperature T_f . Tf Studio simulates that history, analyzes your measurements, and fits relaxation models to them.

Tf Studio contains four tools. They open from the home page (web) or from the launcher (desktop app).

Tool	Use it to...
Learn	understand the experiments (annealing, Kovacs jumps, DSC, enthalpy recovery, MDSC) and the models (TNM, MAP / RelaxPy, Takeda–Lucas), with live simulations on every topic.
Lab	try a cool–heat scan, an anneal or a T-jump interactively and compare the three models side by side.
Fitter	program any thermal history, load your data, then compute with chosen parameters or fit the parameters, with uncertainties.
Data Analysis	turn raw DSC heat flow into normalized C_p , fictive temperatures and recovered enthalpy, and fit volume (or density, length...) relaxation during annealing.

The three models

Model	In one line
Takeda–Lucas (TL)	Heterogeneous domains, each with its own Vogel temperature and local fictive temperature $T_{f,i}$; the spread of $T_{f,i}$ persists in equilibrium, giving temperature-dependent nonexponentiality and enthalpy fluctuation (J. Chem. Phys. 160, 174504, 2024).
Tool–Narayanaswamy–Moynihan (TNM)	One fictive temperature for the whole sample, nonlinearity x and a stretched-exponential (KWW) kernel.
MAP / RelaxPy	Mauro–Allan–Potuzak nonequilibrium viscosity with Prony-series components (Wilkinson, Mauro & Mauro, SoftwareX 2018).

Your data stay on your computer

All calculations run locally, in your browser or in the desktop app. Nothing you load is uploaded anywhere.

Beta software

Tf Studio is in beta: features, defaults and occasionally results may change between versions. Record the version number (bottom of the launcher) with your results.

2 · Getting Tf Studio

2.1 In the browser

Open wtakeda0129.github.io/tf-studio. Nothing to install; it works in Chrome, Edge, Firefox and Safari, and always shows the latest version. After the page has loaded, the tools also work offline.

2.2 Desktop app

Download the installer from github.com/Wtakeda0129/tf-studio/releases/latest, or click **Download for Mac / Windows** on the home page.

macOS (Apple silicon and Intel)

- 1 Download `Tf-Studio-<version>-universal.dmg`.
- 2 Open it and drag **Tf Studio** into **Applications** (choose **Replace** if asked).
- 3 Open Tf Studio. The first time, macOS says it cannot verify the developer: click **Done**.
- 4 Open **System Settings** → **Privacy & Security**, scroll down and click **Open Anyway**, then confirm.

This is needed once per computer, because the app is signed with the group's own certificate rather than an Apple Developer ID.

Windows 10 / 11

- 1 Download `Tf-Studio-Setup-<version>.exe`.
- 2 Run it. If Windows shows “Windows protected your PC”, click **More info** → **Run anyway**.
- 3 Choose the install folder (no administrator rights are needed) and finish.
- 4 Start Tf Studio from the Start menu or the desktop shortcut.

2.3 Updates

The desktop app checks for a new version when it starts and every six hours; you can also check with **Tf Studio** → **Check for Updates...** (Mac) or **Help** → **Check for Updates...** (Windows). A new version downloads in the background and the app offers **Restart now** to install it.

If macOS cannot install an update in place, the app says so and offers **Download**, which opens the release page: install the new DMG as in 2.2. The web version needs no updates.

2.4 The launcher

The desktop app opens a launcher with the four tools; each opens in its own window. Shortcuts: `⌘/Ctrl 1` Learn, `2` Lab, `3` Fitter, `4` Data Analysis, `⌘/Ctrl 0` launcher. The version number is shown at the bottom of the launcher.

3 · Controls shared by every tool

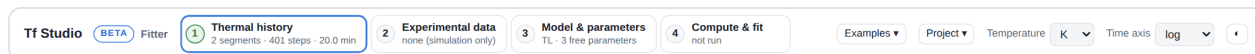


Figure 3.1. Header of the Fitter. The same controls appear in Data Analysis.

Control	What it does
Tf Studio (logo)	Back to the home page or the launcher.
Temperature K / °C	Unit for every temperature on screen: plots, thermal-history inputs, T_g and its bounds, data limits. Values are stored in kelvin, so switching never changes a calculation.
Time axis log / linear	Scale of every time axis (history, relaxation during holds, τ , time-domain data).
☾	Switches between the light (default) and dark colour scheme.
SVG (on each plot)	Saves that plot as a vector graphic for slides or papers.
Hover on a plot	Shows the nearest data point and its values.

Typing numbers

Inputs accept any number format (`1e5` , `0.5` , `-40`). Temperatures are read in the unit selected in the header.

4 · Learn

Learn is a self-contained course on structural relaxation. The contents on the left jump to any section; **Part I** covers experiments and **Part II** the models.

The screenshot shows the Tf Studio Learn interface. At the top, there's a navigation bar with 'Tf Studio BETA Learn' on the left and model selection buttons 'TL', 'TNM', and 'MAP' on the right. Below the navigation bar, the 'CONTENTS' section is visible on the left, listing '1 · Fictive temperature' and 'PART I · EXPERIMENTS' with sub-items like '2 · Isothermal annealing', '3 · Kovacs: asymmetry, memory', '4 · DSC cooling and heating', '5 · Enthalpy recovery', and '6 · MDSC and C_p^* '. The main content area is titled 'Learn: glass relaxation, from experiment to model'. It explains that Part I covers experiments (annealing, Kovacs jumps, DSC, enthalpy recovery, modulated DSC) and Part II covers models (TNM, MAP, Takeda-Lucas). Below this, the section '1 Fictive temperature and structural relaxation' is shown. It describes the fictive temperature T_f and the thermal fingerprint. A mathematical equation for the normalized heat capacity is provided:
$$C_p^N(T) = \frac{C_p - C_{p,g}}{C_{p,l} - C_{p,g}} = \frac{dT_f}{dT}.$$
 The text also mentions that relaxation towards equilibrium has three signatures: Nonexponential, Nonlinear, and Non-Arrhenius. The Non-Arrhenius section mentions the VFT, Adam-Gibbs, and MYEGA forms.

Figure 4.1. Learn. The switch at the top (TL / TNM / MAP) chooses the model used by all simulations in Part I.

Part I · Experiments

Each experiment has its protocol and equations, then a live simulation of the same example glass ($T_g = 734.5$ K, $m = 35.3$):

- **Isothermal annealing:** relaxation function $\phi(t)$ after a 10 K down-jump, compared with a linear KWW curve, and the apparent KWW parameters.
- **Kovacs experiments:** asymmetry of approach (± 5 K jumps) and the memory effect (Kovacs; Macedo–Napolitano crossover).
- **DSC on cooling and heating:** heating scans after five cooling rates and Moynihan's cooling-rate plot (apparent $\Delta h/R$ and fragility).
- **Enthalpy recovery:** heating scans after annealing and $\Delta H(t_a)$.
- **MDSC:** total, reversing and non-reversing signals, C'' , and how the reversing step moves with the modulation period (the Hutchinson and Schawe caveats).

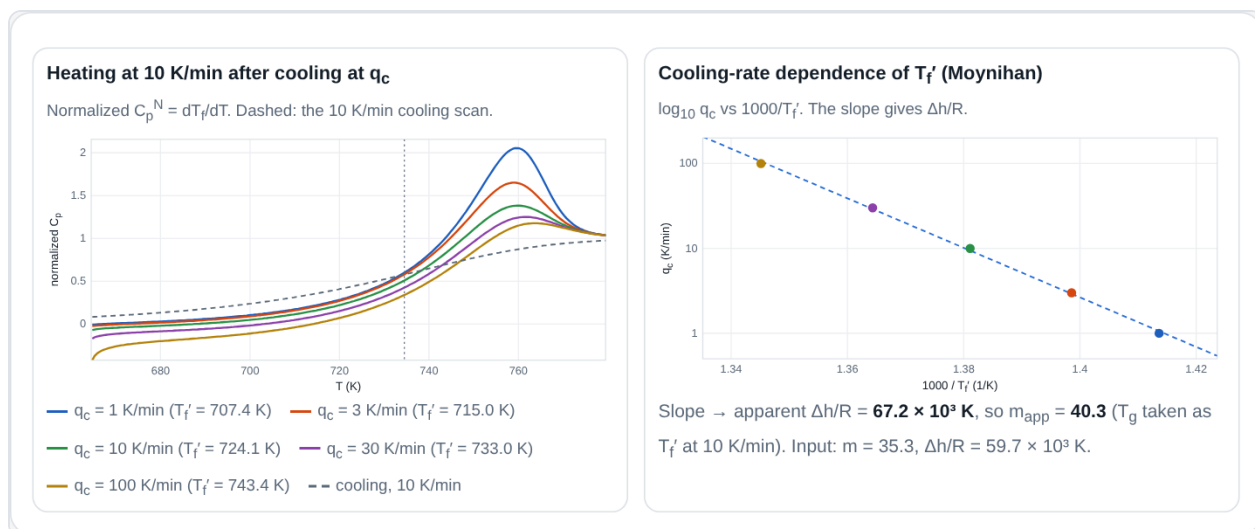


Figure 4.2. DSC section: heating after five cooling rates (left) and the cooling-rate dependence of T_f' (right).



Figure 4.3. MDSC section. Choose **Direction** (heating or cooling) and **Period**; the right plot compares the reversing C_p for three periods.

Part II • Models

The mathematics of TNM, MAP and TL, what each assumes, and an interactive comparison of where the fictive temperatures sit.

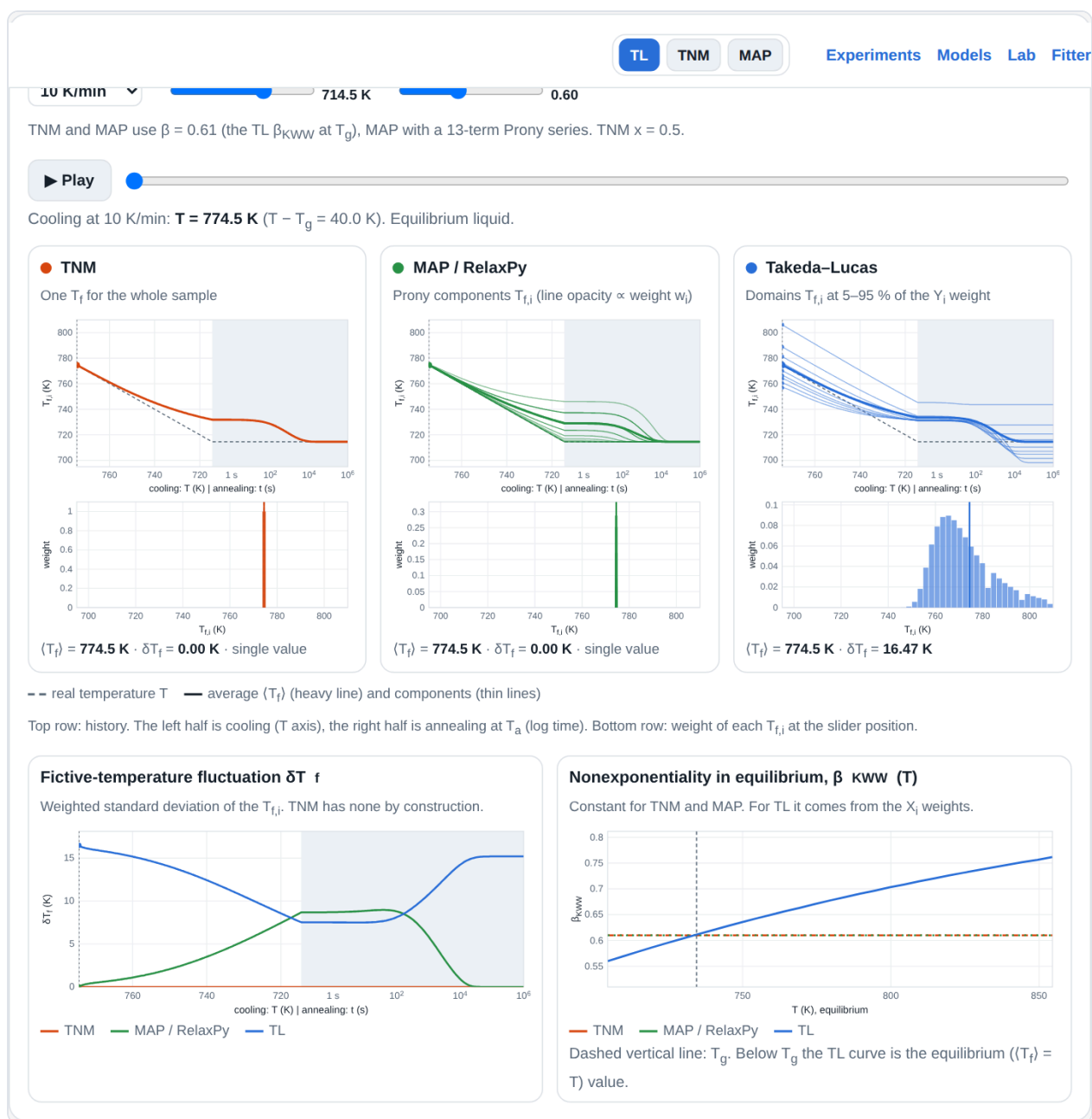


Figure 4.4. “See the difference”: the three models run the same cool–anneal history. Drag the slider or press ► **Play**; change the cooling rate, annealing temperature and TL width f .

5 · Lab

The Lab (formerly Explorer) is a quick, always-live playground: change a number and the plots update. Use it to build intuition; use the Fitter for your own histories and data.

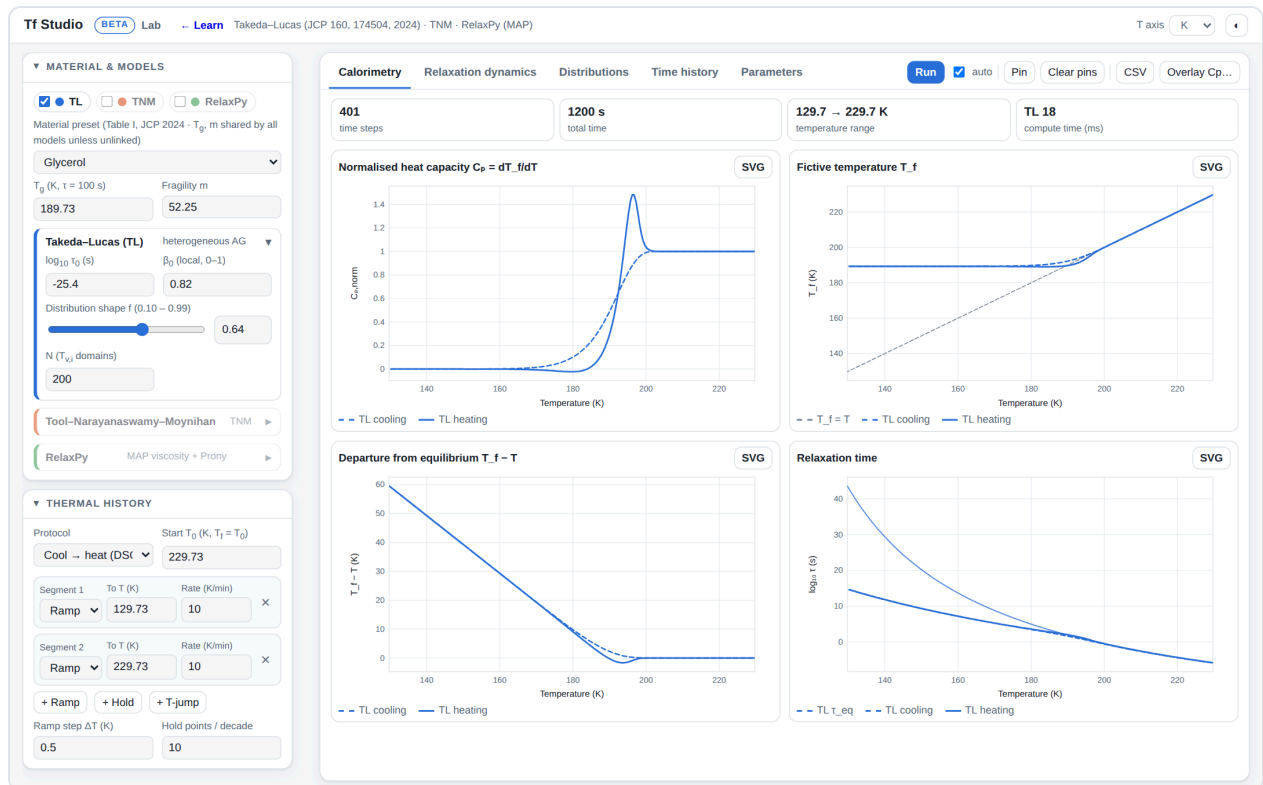


Figure 5.1. Lab: models and material on the left, protocol below; tabs on the right.

- **Models:** tick TL, TNM and/or RelaxPy. **Material preset** loads the TL parameters of Table I of the JCP 2024 paper; T_g and m are shared by the models unless unlinked in their sections.
- **Thermal history:** choose a protocol (cool → heat, slow cool, anneal, Kovacs T-jump) or build segments with **+ Ramp**, **+ Hold**, **+ T-jump**.
- **Tabs:** Calorimetry (C_p , T_f), Relaxation dynamics, Distributions (TL relaxation-time distributions and $\beta_{KWW}(T)$), Time history, Parameters.
- **Run / auto:** untick **auto** to update only when you click **Run**. **Pin** keeps a run for comparison; **CSV** exports the curves.

6 · Fitter

The Fitter works in four steps, shown as buttons across the top. You can move between them at any time.

Step	What you do	Right panel shows
1 Thermal history	Build the temperature program.	T(t) and the time steps
2 Experimental data	Optional: add measurements.	your data and the linked segment
3 Model & parameters	Choose a model, values, which parameters are free, and bounds.	history and data
4 Compute & fit	Click Compute or Run fit .	all results

Nothing is calculated until you click

Steps 1–3 only show the thermal history and your data. The model runs when you click **Compute** or **Run fit** in step 4. Any later change clears the old result and asks you to compute again, so a result on screen always matches the current settings.

6.1 Step 1 · Thermal history

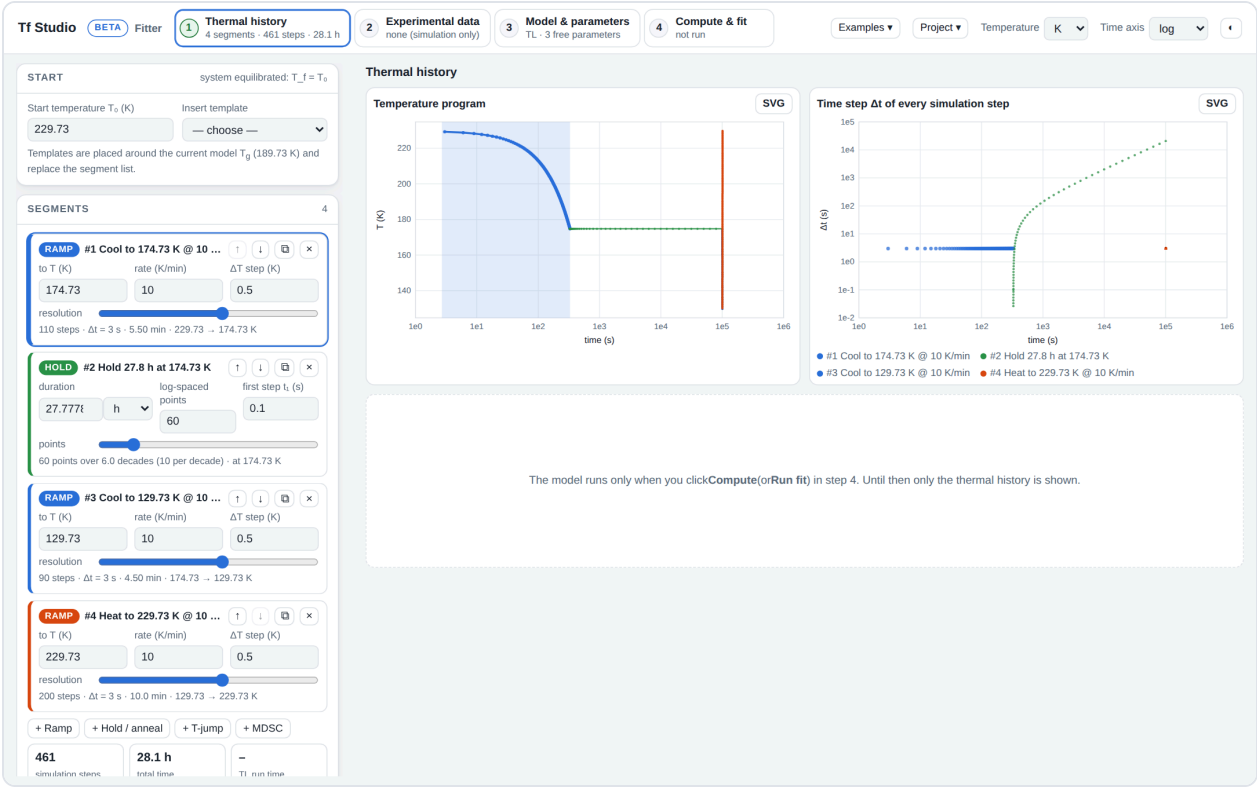


Figure 6.1. Step 1 with the template “Aging below T_g , then DSC heating”.

- 1 **Start temperature T_0 .** The sample starts in equilibrium at T_0 ($T_f = T_0$). Choose T_0 above T_g .
- 2 **Insert template** (optional). Templates are placed around the current model T_g , so set T_g in step 3 first if your material differs from the default.
- 3 **Add segments** with **+ Ramp**, **+ Hold / anneal**, **+ T-jump**, **+ MDSC**. Reorder, duplicate or delete them with the small buttons on each card.

Segment	Inputs	Notes
Ramp	target T, rate (K/min), ΔT step	0.1–0.5 K steps are accurate for DSC; the slider sets ΔT on a log scale.
Hold / anneal	duration (s, min, h, d), log-spaced points, first step t_1	Points are spaced logarithmically from t_1 to the duration.
T-jump	target T	Instantaneous change (e.g. Kovacs experiments).
MDSC	target T, underlying rate, amplitude $\pm A$, period, points per period	Rate 0 gives quasi-isothermal MDSC with a duration.

Under the segments, the summary shows the number of simulation steps and the total time. The right panel plots the temperature program and the time step of every simulation step.

6.2 Step 2 · Experimental data (optional)

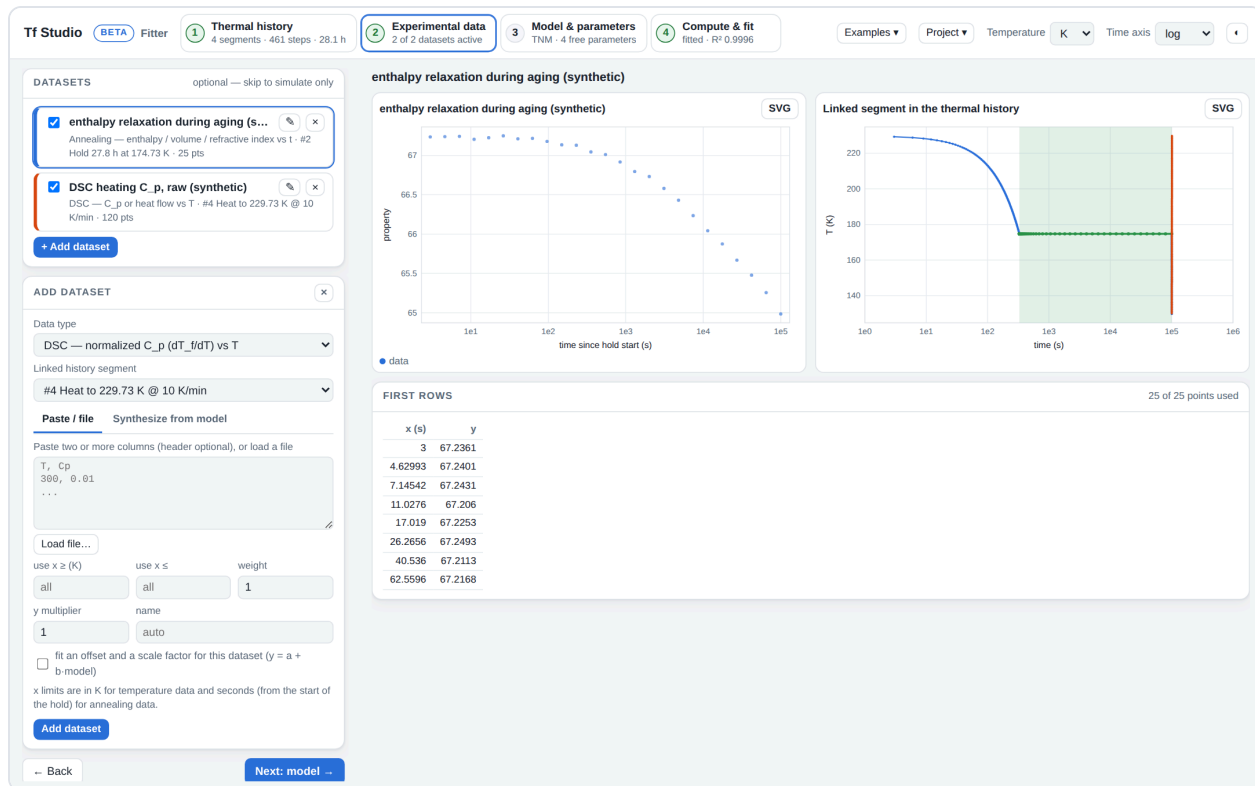


Figure 6.2. Step 2 with two datasets and the **Add dataset** form open.

- 1 Click **+ Add dataset** and choose the **Data type** (table in section 10).
- 2 Choose the **Linked history segment**: the part of the thermal history the data belong to (e.g. the heating ramp for a DSC heating scan, the hold for annealing data).
- 3 **Paste / file**: paste two or more columns or click **Load file...**, then pick the x and y columns and units. Or **Synthesize from model** to generate test data with noise.
- 4 Optional: **use $x \geq$ / $\leq$** to limit the fitted range, a **weight**, a **y multiplier**, and **fit an offset and a scale factor** ($y = a + b \cdot \text{model}$) when the data are not normalized.
- 5 Click **Add dataset**. Tick or untick a dataset to include it; edits it, removes it.

Raw data are fine

Types such as “C_p or heat flow vs T”, “Enthalpy vs T” and “property vs t” fit their baselines or linear coefficients exactly at every step, so you can load raw measurements. To normalize DSC data yourself first, use Data Analysis (section 7).

6.3 Step 3 · Model and parameters

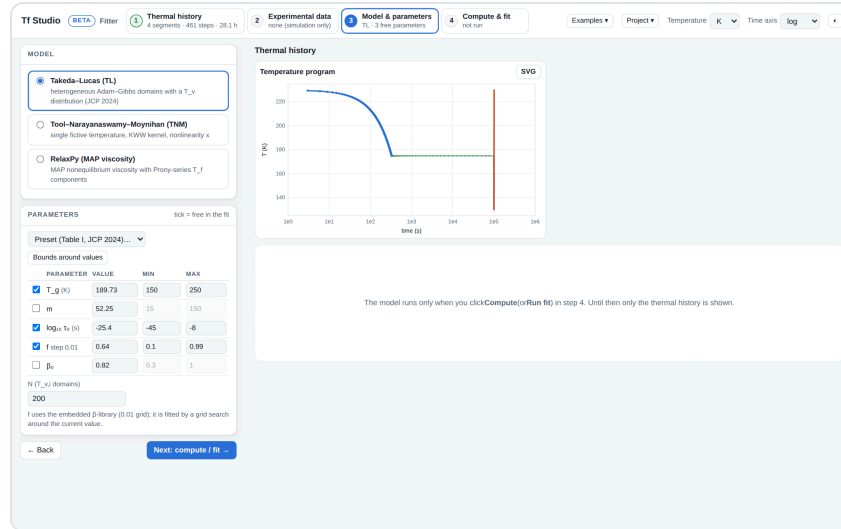


Figure 6.3. Step 3: model, parameter values, tick = free in the fit, and fit bounds.

- Choose **Takeda–Lucas**, **TNM** or **RelaxPy (MAP)**.
- Type values. Ticked parameters are **free** in a fit and move between **min** and **max**; unticked ones stay fixed. **Bounds around values** sets sensible bounds.
- TL: **Preset** loads Table I of the JCP 2024 paper; **N** is the number of domains (200 is accurate). *f* is fitted on its 0.01 grid.
- TNM: choose the exact KWW summation or the Prony kernel for display (fits always use Prony).
- MAP: **RelaxPy example glass**; *A* can follow from continuity at T_g , and $p = 0.3082 \cdot m$ (RelaxPy default).

6.4 Step 4 · Compute and fit

Compute with chosen parameters

Forward calculation, no fitting. Check or adjust the values and click **Compute**. With data, the agreement (R^2 , RMSE per dataset) is shown.

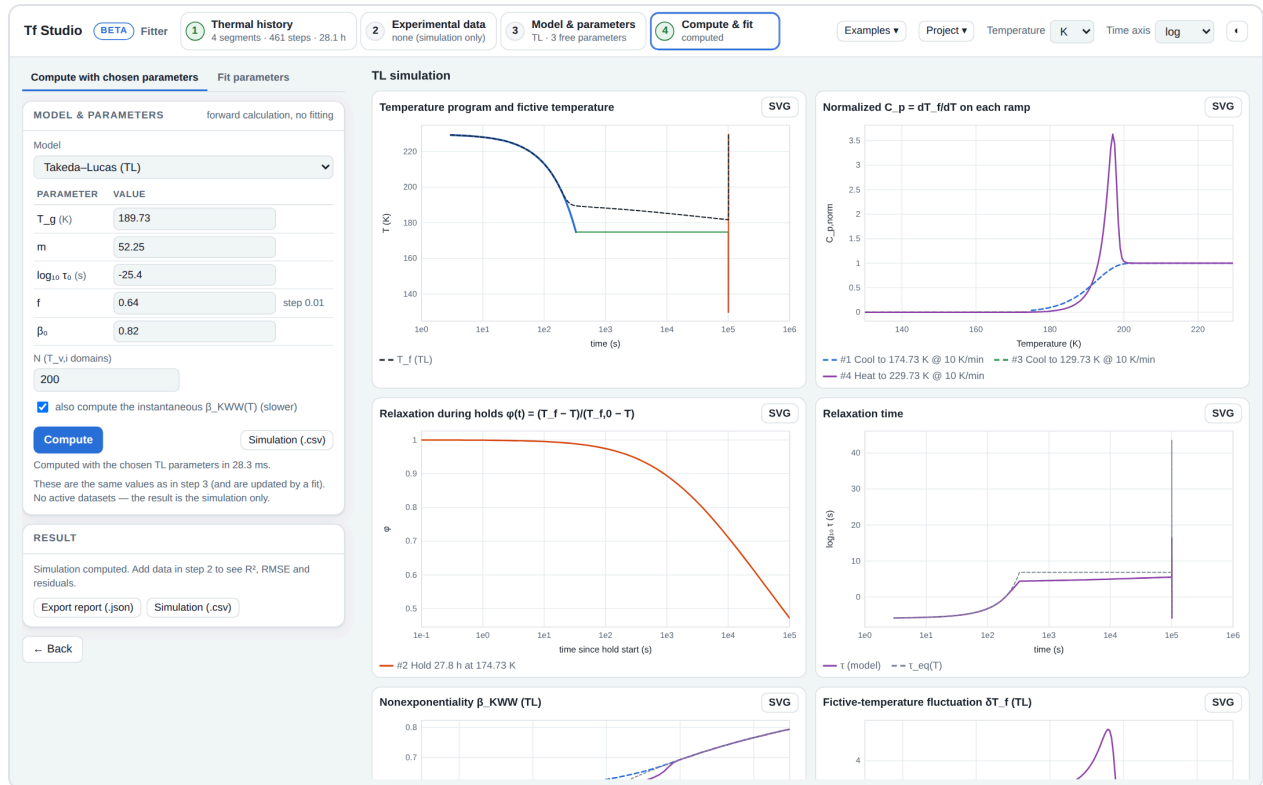


Figure 6.4. Result of **Compute**: temperature program and T_f , normalized C_p on each ramp, relaxation during holds, and relaxation time.

Extra results for the TL model

- **Fictive-temperature fluctuation δT_f vs temperature**, with the equilibrium value.
- **Map of the local fictive temperatures $T_{f,i}$ vs time**, rebinned on a uniform T_f grid (probability density in K^{-1}), with $T(t)$ and $\langle T_f \rangle(t)$.
- **$\beta_{KWW}(T)$** : tick **also compute the instantaneous $\beta_{KWW}(T)$** (adds about a second).

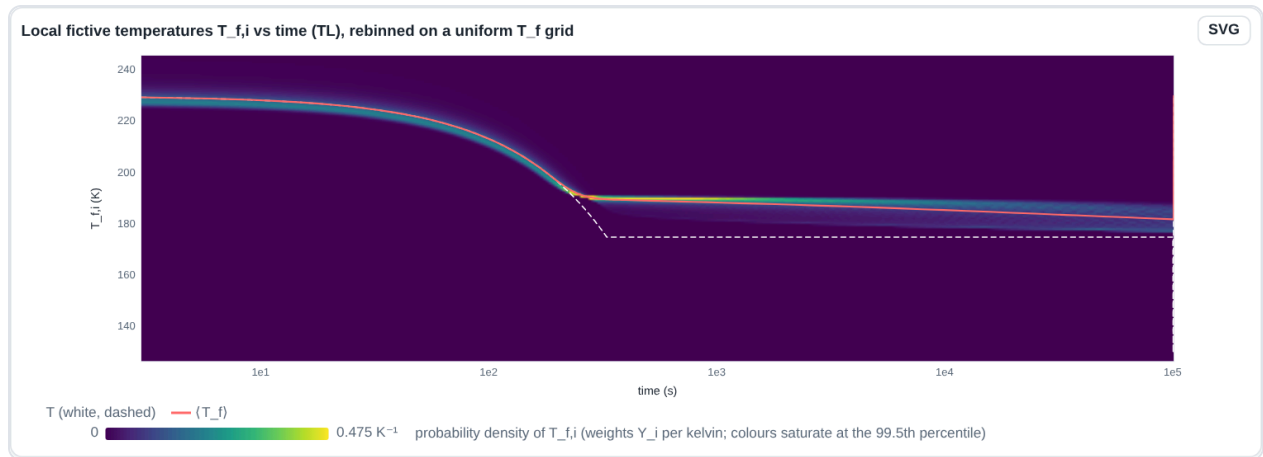


Figure 6.5. TL: local fictive temperatures during cooling, aging and reheating.

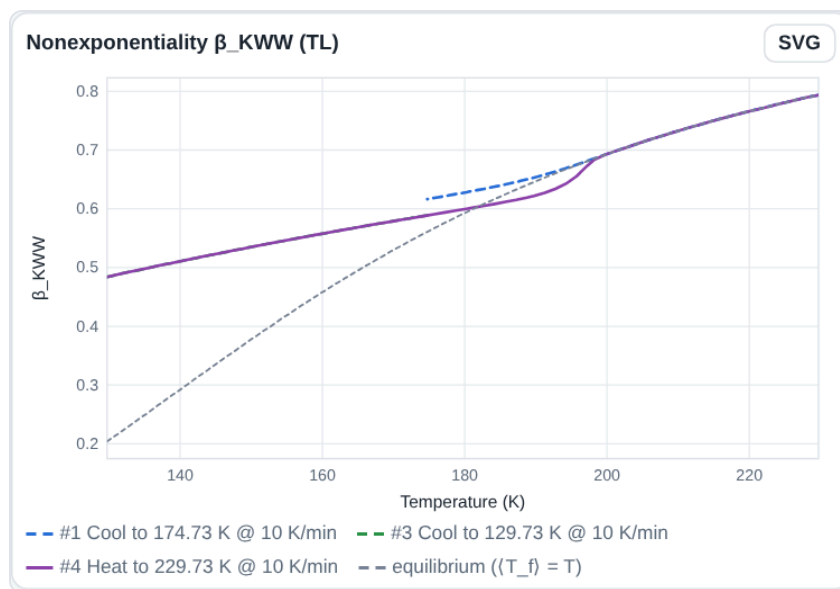


Figure 6.6. TL: instantaneous β_{KWW} along each segment and in equilibrium (grey).

Fit parameters

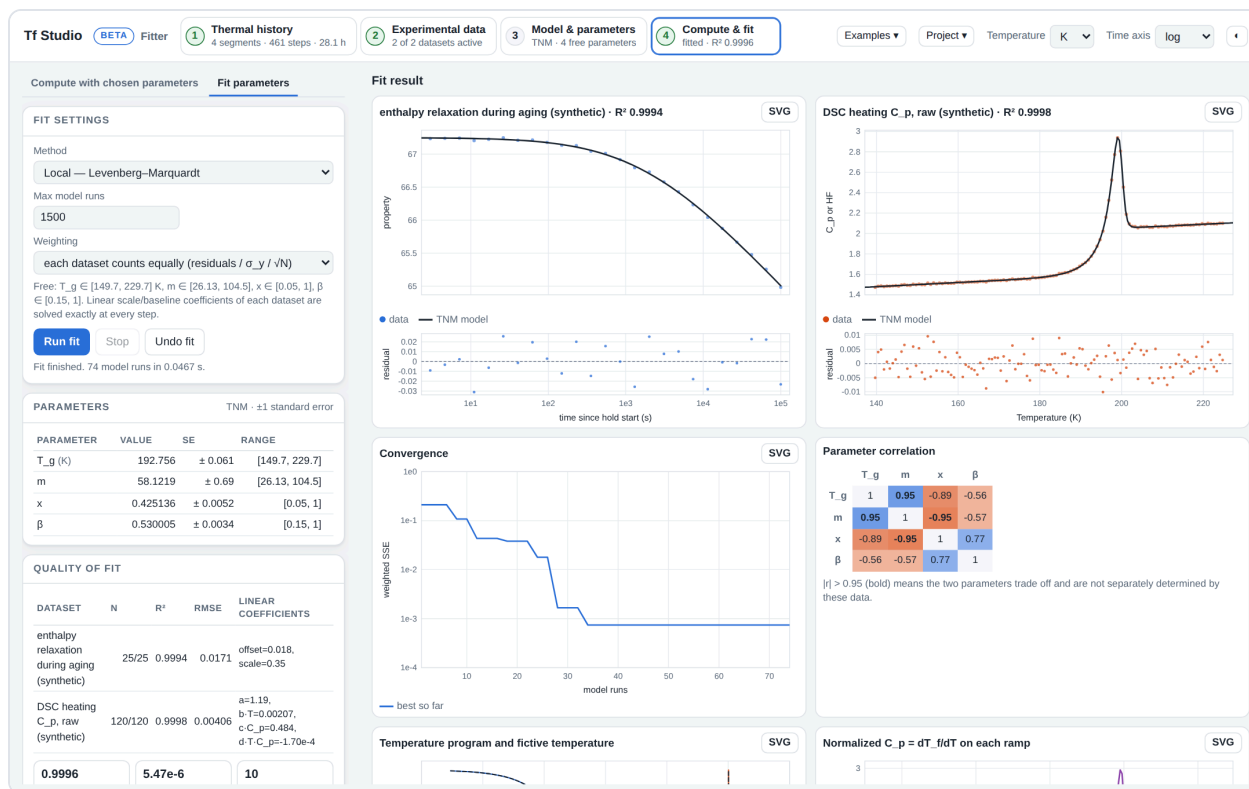


Figure 6.7. Fit result: data with the fitted model and residuals, convergence, parameter correlation, and the parameter table with standard errors.

- 1 Open the **Fit parameters** tab (needs at least one active dataset and one free parameter).
- 2 Choose the **Method**: *Local — Levenberg–Marquardt* (fast, starts from the current values) or *Global — differential evolution + LM polish* (slower, searches the whole box).
- 3 Set **Max model runs** and the **Weighting** (each dataset equal, or each point equal).
- 4 Click **Run fit**. **Stop** ends early; **Undo fit** restores the previous values.

The result lists each parameter with ± 1 standard error, flags parameters *at bound* or *poorly determined*, and gives R^2 , reduced χ^2 , AIC and BIC. In the correlation matrix, $|r| > 0.95$ (bold) means two parameters trade off and are not separately determined by your data.

Exports

Export report (.json) (parameters, errors, statistics, history), **Model vs data (.csv)**, **Simulation (.csv)** (t, T, T_f , τ , τ_{eq} , segment), and the **SVG** button on each plot.

6.5 Projects and examples

Project ▼: *New project*, *Save project (.json)* (history, data, model, values, bounds, fit options) and *Open project....* **Examples ▼** loads ready-made cases: a glycerol DSC cycle, aging + DSC with synthetic enthalpy data, and MDSC with synthetic C_p'/C_p'' .

7 · Data Analysis

Data Analysis prepares your own measurements. It has three tabs across the top. Each tab has **Load example data** (synthetic glycerol data made with the TL model) so you can try everything first.

7.1 Heat-flow scans

ADD A HEAT-FLOW RUN

File (.csv, .txt, instrument export)

Choose File No file chosen

...or paste columns

```
Time (min),Temperature (°C),Heat Flow (W/g)
0.00,-120.0,-0.1000
0.05,-119.5,-0.1000
0.10,-119.0,-0.1000
0.15,-118.5,-0.1000
...
```

161 numeric rows, 3 columns detected.

Temperature column	Heat-flow column
Temperature (°C) ▼	Heat Flow (W/g) ▼
Temperature unit	Heat-flow unit
°C ▼	W/g ▼
Sign convention	Scan
endothermic down (exc) ▼	auto (heating) ▼
Rate q (K/min)	Annealing time t_a (s)
10	none
Annealing T_a (K)	
—	
Name	
auto	
Add run	Cancel

Figure 7.1. Adding a run: paste or load a file, then check the detected columns and units.

- 1 Click **+ Add run**, then choose a file (.csv, .txt, instrument export) or paste columns.
- 2 Check the **Temperature** and **Heat-flow** columns and their units (K or °C; mW, W/g or arbitrary). Column names like "Temperature (°C)" and "Heat Flow (W/g)" set these automatically.
- 3 Set the **Sign convention**: *endothermic up*, or *endothermic down (exo up)* as in TA Instruments exports.
- 4 **Scan** is detected from the temperature trend (heating/cooling). Enter the **rate** and, for mW, the **sample mass**.
- 5 For a heating scan measured after annealing, enter the **annealing time** t_a and T_a . Leave them empty for the unaged reference.
- 6 Click **Add run**.

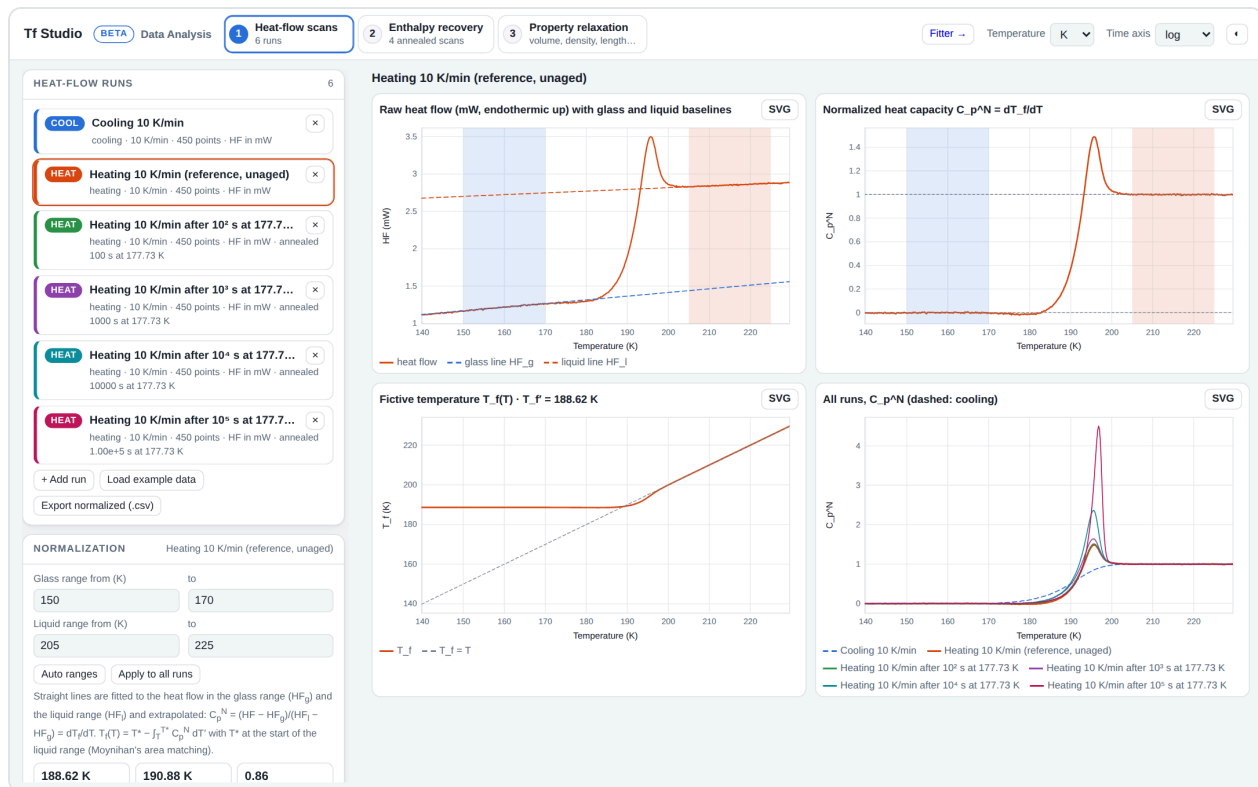


Figure 7.2. Normalization of the reference heating scan. Blue band: glass range; orange band: liquid range; dashed: the fitted baselines.

Normalization. Enter a **glass range** and a **liquid range** (or **Auto ranges**). A straight line is fitted to the heat flow in each range and extrapolated, giving

$$C_p^N = (HF - HF_g) / (HF_l - HF_g) = dT_f/dT, \quad T_f(T) = T^* - \int_T^{T^*} C_p^N dT'$$

with T^* at the start of the liquid range (Moynihan's area matching). The card reports the limiting fictive temperature T_f' , the **midpoint** ($C_p^N = 0.5$) and ΔC_p at T_f' . **Apply to all runs** copies the ranges; **Export normalized (.csv)** saves T , HF , C_p^N , T_f for every run.

Choosing the ranges

- Glass range: well below the transition, where the scan is straight; avoid the start-up transient.
- Liquid range: above the overshoot, where the scan is straight again.
- For comparisons (cooling rates, aging series), use the same ranges for all runs.

7.2 Enthalpy recovery

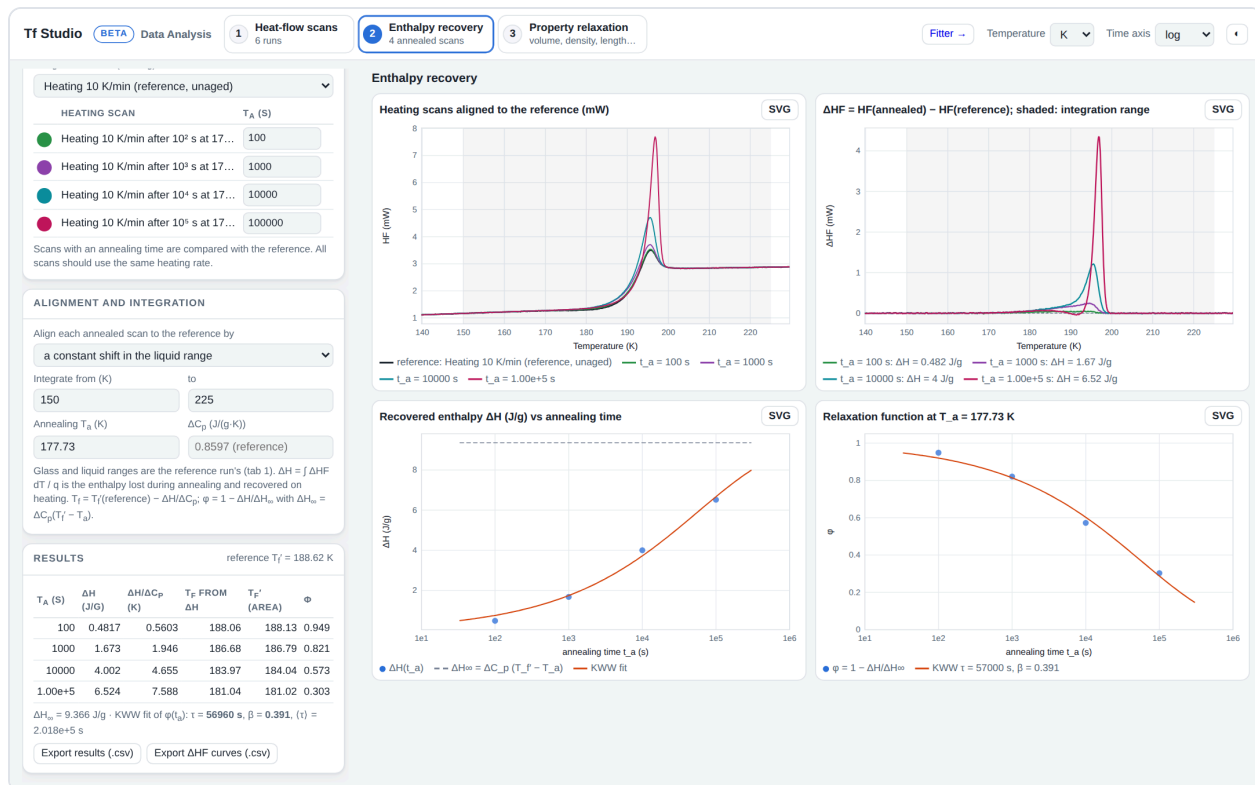


Figure 7.3. Enthalpy recovery: aligned scans, ΔHF(T), recovered ΔH vs annealing time, and φ(t_a) with a KWW fit.

- 1 Choose the **Unaged reference** (a heating scan without annealing).
- 2 Enter or check the annealing time **t_a** of each annealed heating scan.
- 3 Choose how each annealed scan is **aligned** to the reference: a constant shift in the liquid range, or a straight line through the glass and liquid ranges (use this if the glass lines differ slightly).
- 4 Check the **integration range** (default: from the glass range to the liquid range) and enter **T_a**.
- 5 Optionally override **ΔC_p** (default: the reference at T_f').

$$\Delta H = (1/q) \int \Delta HF \, dT, \quad T_f = T_f'(\text{ref}) - \Delta H / \Delta C_p, \quad \phi = 1 - \Delta H / \Delta H_{\infty}, \quad \Delta H_{\infty} = \Delta C_p (T_f' - T_a)$$

Result column	Meaning
ΔH	enthalpy lost during annealing and recovered on heating (J/g with mW + mass or W/g; mJ with mW and no mass)
$\Delta H/\Delta C_p$	the same in kelvin: the drop of the fictive temperature during annealing
T_f from $\Delta H / T_f'$ (area)	fictive temperature after annealing, from ΔH and independently from area matching of the annealed scan (they should agree)
φ	relaxation function; with three or more annealing times a KWW fit gives τ , β and $\langle \tau \rangle$

Use the same heating rate

ΔH is computed with the reference's rate. All scans should be heated at the same rate and ideally measured on the same sample.

7.3 Property relaxation

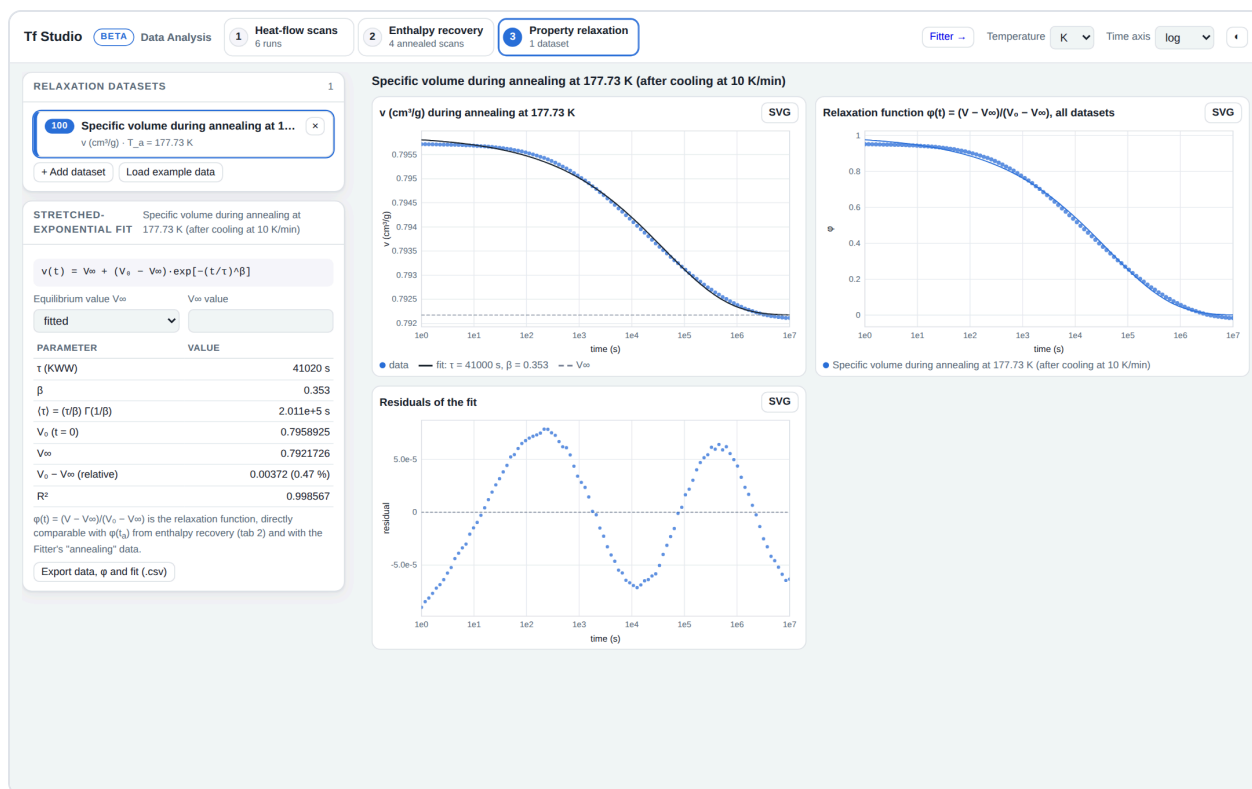


Figure 7.4. Specific volume during annealing with the stretched-exponential fit, $\varphi(t)$ and residuals.

Add a dataset of a property against time during isothermal annealing: volume, density, length, refractive index... (+ **Add dataset**: pick the time column and unit, the property column, a label and T_a). The fit is

$$V(t) = V_\infty + (V_0 - V_\infty) \exp[-(t/\tau)^\beta]$$

with V_∞ **fitted** or **fixed** at a value you enter (use fixed when the data stop well before equilibrium). The table gives τ , β , $\langle \tau \rangle = (\tau/\beta)\Gamma(1/\beta)$, V_0 , V_∞ , the relative change and R^2 . Several datasets can be compared on the $\phi(t)$ plot.

Systematic residuals (as in Figure 7.4) are expected for real glasses: relaxation is nonlinear, so a single stretched exponential is only an approximation. Fitting the model in the Fitter accounts for this.

7.4 From analysis to fit

Export the normalized curves (tab 1) or $\phi(t)$ (tabs 2, 3) as CSV and paste them into Fitter step 2 as “DSC — normalized C_p ” or “Annealing — normalized relaxation $\phi(t)$ ”, linked to the matching segment of a thermal history that reproduces your experiment.

8 · Data files

- **Formats:** text files (.csv, .txt, .dat, .tsv) or pasted text. Columns may be separated by commas, semicolons, tabs or spaces.
- **Headers:** lines before the numbers (titles, column names, unit rows) are skipped; the last text line before the data is used for column names.
- **Decimal point:** use a period (0.25). Files with decimal commas must be converted first.
- **Order:** any; cooling scans may run from high to low temperature.
- **Instrument exports:** export to text/CSV from the instrument software (e.g. TA Universal Analysis / TRIOS, Netzsch Proteus, Mettler STARE) with temperature and heat flow columns.

Example (DSC, TA style)	Example (annealing)
Time (min),Temperature (°C),Heat Flow (W/g)	t (s) V (cm3/g)
0.00, -120.0, -0.1000	1 0.795716
0.05, -119.5, -0.1001	1.2 0.795715
...	...

9 · Troubleshooting and questions

Problem	What to do
The Fitter shows no model curves.	Expected: the model runs only when you click Compute or Run fit in step 4.
A plot shows “Could not draw ...”.	The rest of the page still works. Check the inputs of that part (e.g. a range with fewer than two points) and report it if it persists.
A hold shows no relaxation.	The hold is above T_g : the liquid equilibrates within milliseconds. Put the hold below T_g .
Mac: “cannot be opened / cannot verify the developer”.	System Settings → Privacy & Security → Open Anyway (once per computer).
Mac: “Restart now” does nothing, or the version does not change.	Download the latest DMG and replace the app (section 2.2). Versions before 1.1.6 cannot update themselves.
Windows: “Windows protected your PC”.	More info → Run anyway (first install only).
TL map or β_{KWW} plot missing.	Very long histories (above ~60 000 steps with 200 domains) do not keep every $T_{f,i}$. Use a coarser ΔT step or fewer hold points.
A fitted parameter is “at bound”.	Widen its bounds in step 3, or fix it if the data cannot determine it.
Parameters strongly correlated ($ r > 0.95$).	Add data that separate them (several cooling rates, annealing data) or fix one of them.
C_p^N is upside down.	Change the sign convention of the run (endothermic up/down).
ΔH has the wrong sign or size.	Check the sign convention, the rate and the sample mass, and that all scans used the same heating rate.

Report problems or ideas at github.com/Wtakeda0129/tf-studio/issues or to the Lucas group.

10 · Reference

Model parameters

Model	Parameter	Meaning
TL	T_g	temperature where $\langle \tau \rangle = 100$ s
	m	fragility index
	$\log_{10} \tau_0$	high-temperature limit of τ (s)
	f	width of the relaxation-time distribution ($\approx \beta_{\text{KWW}}$ at T_g), 0.10–0.99
	β_0	intrinsic (local) stretching; 1 = fully heterogeneous
	N	number of domains (numerical; 200 is accurate)
TNM	T_g	$\tau(T_g) = 100$ s
	m	fragility; $\Delta h/R = m \ln 10 T_g$
	x	nonlinearity (0–1)
	β	KWW stretching exponent
MAP / RelaxPy	T_g	$\eta(T_g) = 10^{12}$ Pa·s
	m	fragility (MYEGA)
	$\log_{10} \eta_\infty$	high-temperature viscosity limit
	A, B, C	nonequilibrium viscosity branch; A from continuity at T_g by default
	p	ergodicity exponent (default $0.3082 \cdot m$)
	$\log_{10} K_s$	modulus in $\tau = \eta/K_s$
	β	stretching of the Prony series

Fitter data types

Data type	Linked to	Notes
DSC — normalized C_p (dT_f/dT) vs T	ramp	already normalized (e.g. from Data Analysis)
DSC — C_p or heat flow vs T	ramp	raw; glass/liquid baselines fitted
Enthalpy vs T	ramp, MDSC	$H = a + bT + \Delta C_p T_f$ fitted
Volume / density / length vs T	ramp, MDSC	dilatometry
Fictive temperature T_f vs T	ramp, MDSC	
Annealing — normalized $\varphi(t)$	hold	$\varphi = (T_f - T)/(T_{f,0} - T)$
Annealing — $T_f(t)$	hold	
Annealing — enthalpy / volume / refractive index vs t	hold	$a + b \cdot T_f$ fitted
MDSC — C_p' (reversing), C_p''	MDSC	normalized; optional scale

Glossary

T_f	fictive temperature: the temperature at which the liquid would have the current structure
T_f'	limiting fictive temperature of the glass (frozen value after cooling)
C_p^N	normalized heat capacity, 0 in the glass and 1 in the liquid; equals dT_f/dT
$\varphi(t)$	normalized relaxation function, $1 \rightarrow 0$ during annealing
KWW	stretched exponential $\exp[-(t/\tau)^\beta]$
δT_f	spread (standard deviation) of the local fictive temperatures in the TL model

11 · How to cite, and references

If Tf Studio contributes to your work, please cite:

- W. Takeda and P. Lucas, “A model of heterogeneous undercooled liquid and glass accounting for temperature-dependent nonexponentiality and enthalpy fluctuation,” *J. Chem. Phys.* **160**, 174504 (2024). doi:10.1063/5.0196812
- W. Takeda and P. Lucas, “Relationship between enthalpy fluctuation and nonexponential relaxation in glass-forming liquids,” *Mater. Adv.* **7**, 5729 (2026). doi:10.1039/d6ma00442c

and, when you use them, the TNM and MAP / RelaxPy papers:

- A. Q. Tool, *J. Am. Ceram. Soc.* **29**, 240 (1946); O. S. Narayanaswamy, *J. Am. Ceram. Soc.* **54**, 491 (1971); C. T. Moynihan et al., *J. Am. Ceram. Soc.* **59**, 12 (1976); I. M. Hodge, *Macromolecules* **20**, 2897 (1987); G. W. Scherer, *J. Am. Ceram. Soc.* **69**, 374 (1986).
- J. C. Mauro, D. C. Allan and M. Potuzak, *Phys. Rev. B* **80**, 094204 (2009); X. Guo et al., *J. Am. Ceram. Soc.* **101**, 1169 (2018); J. C. Mauro and Y. Z. Mauro, *Physica A* **506**, 75 (2018); J. C. Mauro, R. J. Loucks and P. K. Gupta, *J. Am. Ceram. Soc.* **92**, 75 (2009); C. J. Wilkinson, Y. Z. Mauro and J. C. Mauro, *SoftwareX* **7**, 255 (2018).

Experiment references (Kovacs, Macedo–Napolitano, Hodge, Schawe, Hutchinson–Montserrat) are listed in Learn → Further reading.

Tf Studio · Lucas group, Department of Materials Science and Engineering, University of Arizona · github.com/Wtakeda0129/tf-studio